

Group: _____ **Section:** _____

Member 1: _____ **BUID:** _____

Member 2: _____ **BUID:** _____

Member 3: _____ **BUID:** _____

Member 4: _____ **BUID:** _____

Project 2 Simulation Design

Purpose

Today you will turn the Project 2 plan into a simulation algorithm that compares classical and bootstrap inference across planned scenarios.

Shared Group Workflow

Work in one shared Google Doc and one shared analysis folder. Assign the four roles below, then have every member review the simulation design and outputs.

- **Member 1 – Scenario 1 lead:** define and code the first violation scenario.
- **Member 2 – Scenario 2 lead:** define and code the second violation scenario.
- **Member 3 – Methods lead:** implement classical and bootstrap comparisons and coverage measures.
- **Member 4 – Results editor:** organize tables, plots, reproducibility notes, and the shared report.

In-Class Work

- (A) **Define scenarios.** Specify the data-generating process, assumptions being violated, and parameter values.
- (B) **Write the algorithm.** Build code for one simulated data set, then wrap it in repeated simulation.
- (C) **Compare methods.** Store classical intervals, bootstrap intervals, coverage indicators, and interval widths.
- (D) **Output plan.** Decide which tables and plots will summarize the simulation results.
- (E) **Check-in.** How will you know whether one inference method is performing better?

Deliverable

Submit the Project 2 simulation-design materials through Gradescope or Blackboard by Sunday, October 25 at 10:00 PM. Include the simulation algorithm, planned outputs, shared Google Doc link, and reproducible code files.